

Slate User Manual

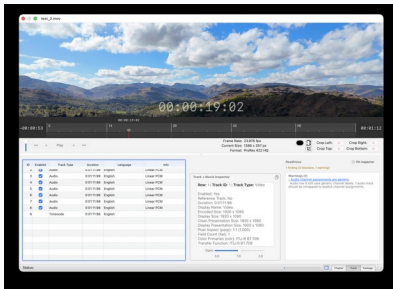
Media package inspection and readiness review for macOS

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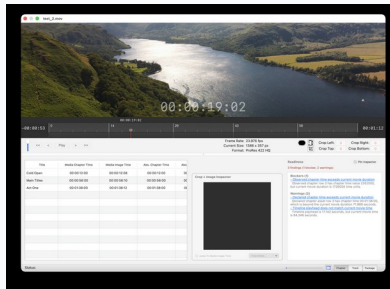
Open. Inspect. Compare. Act.

Slate helps media operations teams inspect movies, package metadata, tracks, chapters, sidecars, crop values, and readiness findings before delivery. Open a movie, open its package, and Slate compares declared package data against observed media state so operators can catch blockers and warnings earlier.

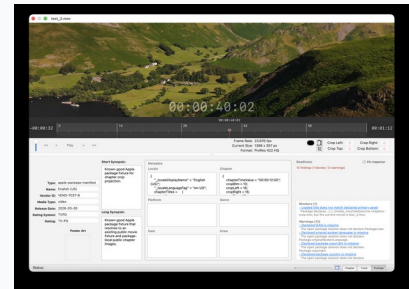
Track View



Chapter View



Package View



Use Slate to

Validate package metadata against the opened movie.
Inspect audio, video, subtitle, caption, and reference tracks.
Review chapters, chapter images, crop values, and sidecars.
Run readiness checks from the app or command line.

Result

Find delivery blockers before handoff.
Review media state in one operator workspace.
Resolve package and content issues faster.
Support both human review and scripted validation.

Contents

- 1. Why Slate
- 2. Before You Begin
- 3. Quick Start
- 4. Readiness Concepts
- 5. Visual Workspace Overview
- 6. App Workflows
- 7. CLI Workflows and Sample Output
- 8. Menu and Shortcut Reference
- 9. Troubleshooting
- 10. Operator Notes
- Appendix A. Public Application Build Notes
- Glossary

1. Why Slate

Slate is a macOS inspection and readiness-review tool for media packages. It lets operators open a movie, open a package manifest, inspect tracks, compare declared package data against observed media state, preview chapter metadata and images, and review readiness blockers and warnings before delivery.

Audience	How Slate helps
Media operations teams Package validation teams	Review movies, packages, and sidecars before delivery handoff. Compare declared package metadata with observed movie/session state.
Localization and QC operators	Inspect tracks, languages, captions, subtitles, chapters, and chapter images.
Automation and build teams	Use CLI output for scripted checks, CI jobs, and archival reports.

Inspection, not editing

Slate is primarily an inspection and validation application. Reference tracks, sidecars, crop overlays, and pane selections are session tools that help you inspect and validate the media you have opened. They do not edit the package manifest.

2. Before You Begin

Input type	Supported inputs	Notes
Movies	.mov, .mp4	Use the movie declared by the package primary video asset when possible.
Packages	Package JSON files; Apple manifest XML files	Package and movie data work best together when the opened movie matches the package declaration.
Reference audio/video tracks	.mov, .wav, .aif, .aiff, .m4a, .mp3, .caf	Reference tracks are temporary session tools for inspection and playback comparison.
Subtitle and caption sidecars	.itt, .srt, .scc, .ttml, .dfxp	Sidecars can be added manually or autoloaded when they sit next to the movie and share its base filename.

Best results checklist

1. Place the package JSON or Apple manifest XML where Slate can access it.
2. Open the movie that matches the package declared primary video asset.
3. Place subtitle or caption sidecars next to the movie when using autoload.
4. Open the Readiness Report after both package and movie are loaded.

3. Quick Start

1. Open a movie with File > Open Movie... or press Command-O.
2. Open a package with File > Open Package... or press Shift-Command-P.
3. Use View > Tracks, View > Chapters, or View > Package to move through the workspace panes.
4. Use Validation > Show Readiness Report to view the full readiness result.
5. Click readiness links in the pane rails to jump to the relevant row or field.

Drag-and-drop shortcut

You can also drag a .mov file directly onto the viewer to open it.

4. Readiness Concepts

Slate readiness checks combine package data and observed movie/session state when both are available. When a package is open but no movie is loaded, Slate can still inspect package structure, but declared-vs-observed media checks are limited.

Status	Meaning	Operator action
Pass	No current blockers or warnings.	Continue review or delivery workflow.
Warning	Delivery may need review.	Inspect the finding, confirm whether it is acceptable, and resolve where needed.
Blocker	A condition should be fixed before continuing.	Stop handoff until the blocker is resolved or intentionally waived by your process.

Readiness area	Where it appears	What it covers
Package	Package View readiness rail	Package metadata, platform data, synopsis, poster art, locale metadata, asset declarations.
Track	Track View readiness rail	Movie tracks, roles, language values, audio/video state, reference tracks.
Chapter	Chapter View readiness rail	Chapter rows, chapter timing, image status, crop values, movie-aware chapter comparisons.
Full report	Validation > Show Readiness Report	All current findings across the open package and observed session.

5. Visual Workspace Overview

Slate has three primary workspaces. Each pane keeps the viewer, timeline, status area, and readiness rail visible so operators can review media context and validation findings together.

Track View

Inspect the movie and its readiness status in one view. Track View combines playback, timeline navigation, track metadata, sidecar review, inspector details, and readiness findings so operators can diagnose issues without switching tools.

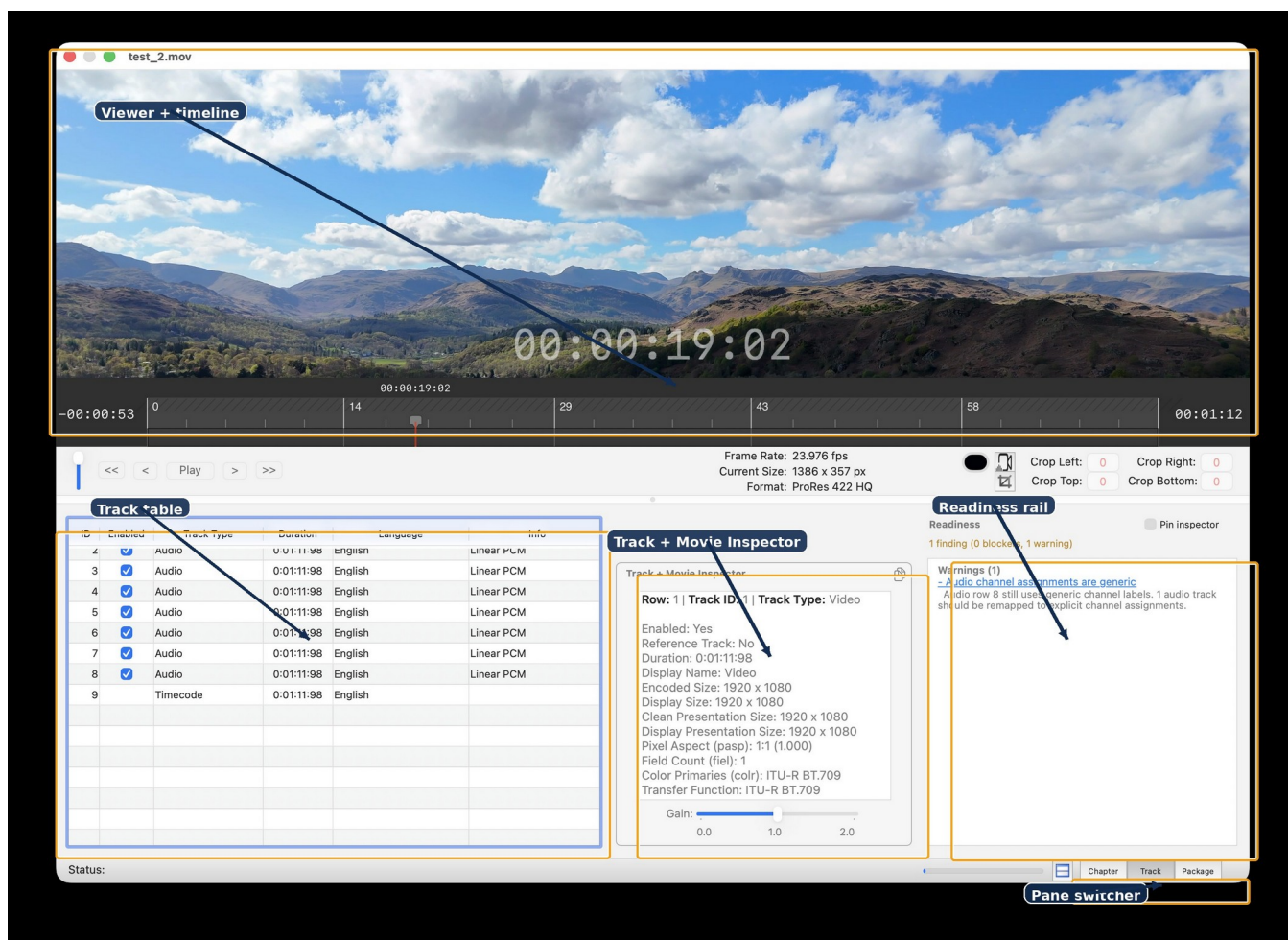


Figure 1. Track View: viewer and timeline, track table, inspector, readiness rail, and pane switcher.

Callout	Use it for
Viewer + timeline	Play the movie, scrub, frame-step, review timecode, and set or inspect crop overlays.
Track table	Review track IDs, enabled state, types, duration, language, and format information.
Track + Movie Inspector	Copy details, inspect row state, and adjust session controls such as audio gain when available.
Readiness rail	Review track and role findings and jump to the relevant row or control.
Pane switcher	Move between Chapter, Track, and Package views.

Chapter View

Review chapter timing and image readiness. Chapter View shows package chapter rows, timing comparisons, crop/image preview state, and chapter-specific findings.

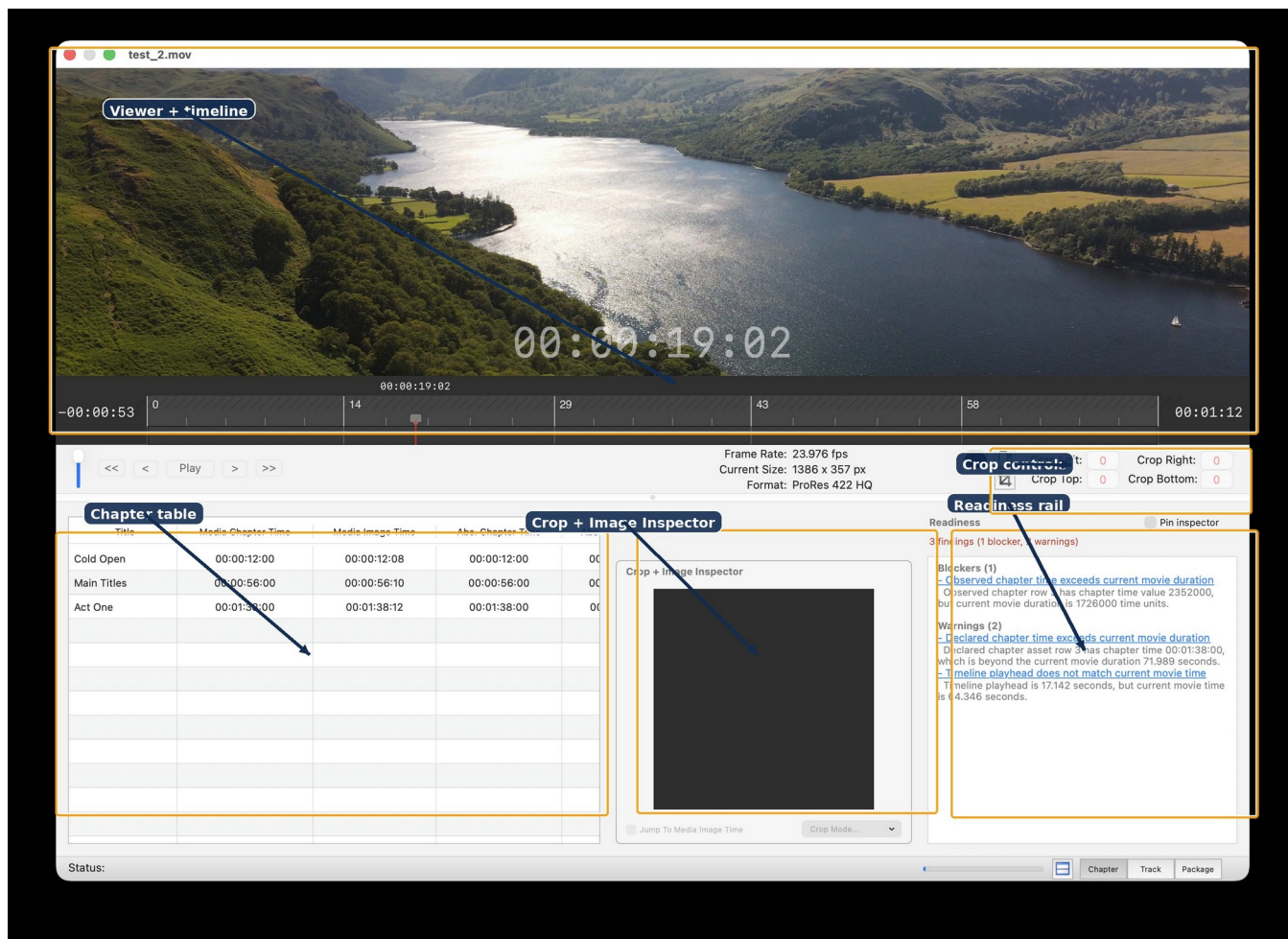


Figure 2. Chapter View: package chapter rows, crop/image inspector, crop controls, and readiness findings.

Callout	Use it for
Chapter table	Review titles, declared chapter times, image times, absolute timing, crop values, and image status.
Crop + Image Inspector	Preview chapter imagery and inspect crop state before capture or evaluation.
Readiness rail	Find chapter blockers or warnings and jump to the associated row.
Crop controls	Review crop left/right/top/bottom values and evaluate crop mode decisions.

Package View

Review declared package metadata and package-level readiness. Package View brings together package identity, synopsis fields, poster art, metadata collections, chapter metadata, asset declarations, and package findings.

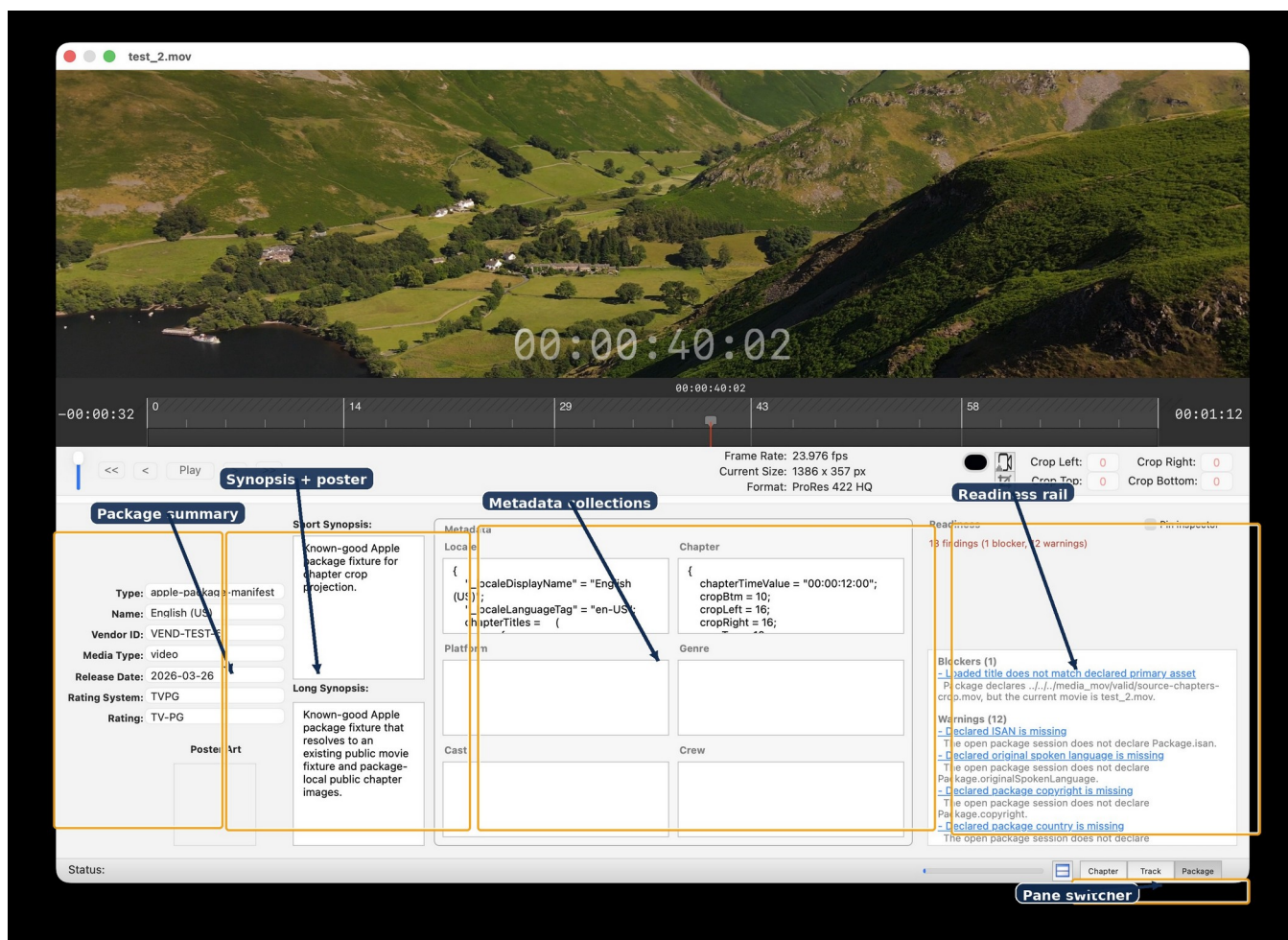


Figure 3. Package View: package summary, synopsis, metadata collections, readiness rail, and pane switcher.

Callout	Use it for
Package summary	Confirm type, name, vendor, media type, release date, rating system, and rating.
Synopsis + poster	Review short synopsis, long synopsis, and poster art state.
Metadata collections	Inspect locales, genres, platforms, cast, crew, and chapter metadata.
Readiness rail	Review package and metadata findings and jump to the relevant field when available.

6. App Workflows

Opening recent inputs

Use File > Open Recent to reopen recent operator inputs. Slate filters this list to supported movies and package files. Use Clear Menu to remove recent entries.

Playback controls

Control	Purpose
<<	Shuttle or fast-reverse playback.
<	Step backward.
Play / Pause	Start or stop playback.
>	Step forward.
>>	Fast-forward playback.
Volume slider	Adjust playback volume.
Spacebar	Toggle play and pause when you are not editing text.

Viewer options and crop overlay

- Use the color well to change the viewer background. Slate remembers the selected background color.
- Use the timecode button to show or hide the timecode overlay when the loaded movie contains a timecode track. If no timecode track is available, the button is disabled.
- Use the crop button, or Command-click in the viewer, to create a crop overlay. Drag inside the overlay to move it, or drag its handles to resize it. Option-click removes the crop overlay.
- Crop values appear in the crop fields and status area.

Track View tasks

Track View lists the tracks in the current movie and any loaded reference tracks. Select a row to update the Track + Movie Inspector.

Column	What it shows
ID	Track identifier.
Enabled	Whether the track is active for supported playback.
Track Type	Audio, video, subtitle, caption, timecode, or reference track type.
Duration	Track duration.
Language	Declared or applied language value.
Info	Format, role, source, or summary details.

Task	How to do it
Resolve unknown asset type	Choose an Asset Type when readiness reports that the asset type is unknown.
Enable or disable playback	Use the Enabled column for supported track rows.
Review details	Select a row to view track ID, media type, duration, language, format summary, source path, and audio gain when available.
Copy details	Use the copy button in the inspector.
Add reference media	Use Track > Add Reference Track...
Delete reference track	Use Track > Delete Reference Track or press Command-D.

Reference tracks and sidecars

- Reference tracks are used for inspection and playback comparison in the current session. They do not edit the package manifest.
- Slate can autoload sidecar files that sit next to the movie and share the movie base filename.

- Supported autoload sidecars are .itt, .srt, .scc, .ttml, and .dfxp.
- Use View > Autoload Sidecars to turn automatic sidecar loading on or off.
- Use View > Show Subtitles or View > Show Closed Captions to show or hide loaded subtitle/caption sidecars. These menu items are enabled only when matching sidecar tracks are present.

Chapter View tasks

Chapter View is package-driven. Open a package before using it. When movie timing is available, selecting a chapter can jump the viewer to the chapter image time.

Column	What it shows
Title	Declared chapter title.
Media Chapter Time	Chapter start time relative to the media.
Media Image Time	Image time relative to the media.
Abs. Chapter Time	Absolute chapter time when available.
Abs. Image Time	Absolute image time when available.
Crop Values	Declared or evaluated crop values.
Image	Chapter image status or preview state.

Crop mode	Behavior
Manual	Uses the crop overlay currently set in the viewer.
Conservative Auto	Detects conservative black-border crop values from sampled frames.
None	Captures or evaluates the full image area.

Package View tasks

Package View shows the opened package declared metadata and assets. Open a package with File > Open Package....

Area	What to inspect
Identity	Type, name, vendor, media type, release date, rating system, and rating.
Assets	Asset stack and declared primary asset information.
Synopsis	Short and long synopsis text.
Collections	Platforms, locales, genres, cast, crew, and chapters.
Poster art	Poster art details and readiness state.
Findings	Package and metadata readiness findings.

Readiness report in the app

- Use Validation > Show Readiness Report or press Option-Command-R to open the full readiness report.
- Use Validation > Show Package Summary or press Option-Command-P to open a package summary dialog. When a validation report is available, the summary includes readiness status and operator-readable findings.
- Clicking a readiness finding jumps to the most relevant field, row, or control when Slate can identify one.
- If a package is open but no movie is loaded, declared-vs-observed media checks are limited. Open the declared primary movie to enable full comparison.

7. CLI Workflows and Sample Output

Slate also includes command-line runtime tools for scripted validation workflows. Use the CLI when you need repeatable checks, machine-readable JSON, CI integration, or a report archive outside the app.

Command	Purpose	Best for
<code>./slate readiness source-chapters-crop.itmsp/metadata.xml</code>	Prints a human-readable readiness summary.	Operator terminal review and quick triage.
<code>./slate analyze --package source-chapters-crop.itmsp/metadata.xml</code>	Prints a JSON array of findings.	Automation, filtering, and CI logic.
<code>./slate report source-chapters-crop.itmsp/metadata.xml</code>	Prints a structured JSON report with summary, operator text, next finding, and payload.	Archival output and systems integration.
<code>./slate analyze --package /path/to/package.json --observed-state /path/to/observed-state.json</code>	Compares package declarations against a supplied observed-state payload outside the app.	Movie-aware checks in scripted workflows.

Sample package used in this section

The examples below use `source-chapters-crop.itmsp/metadata.xml`. The sample readiness result contains 10 warnings, 0 blockers, and a next finding of Metadata / Declared ISAN is missing. It also shows a deferred comparison because no movie was loaded for declared-vs-observed media checks.

Human-readable readiness output

Use readiness output when an operator needs a fast, readable answer in the terminal. It starts with the overall count, shows the next recommended finding, lists warnings or blockers, and then explains deferred comparisons.

```
./slate readiness source-chapters-crop.itmsp/metadata.xml
Readiness: 10 warnings
Summary:
- Blockers: 0
- Warnings: 10

Next:
- Metadata / Declared ISAN is missing

Warnings:
- Metadata: Declared ISAN is missing
  The open package session does not declare Package.isan.
- Metadata: Declared original spoken language is missing
  The open package session does not declare Package.originalSpokenLanguage.
- Metadata: Declared package copyright is missing
  The open package session does not declare Package.copyright.
- Metadata: Declared package country is missing
  The open package session does not declare Package.country.
- Metadata: Declared platform data is missing
  The open package session does not declare any platform-data rows.
- Metadata: Declared production company is missing
  The open package session does not declare Package.productionCompany.
- Metadata: Primary locale cast is missing
  The open package session does not expose cast rows for the primary locale.
- Metadata: Primary locale crew is missing
  The open package session does not expose crew rows for the primary locale.
- Metadata: Primary locale poster art is missing
  The open package session does not expose Locale.posterArt for the primary locale.

Deferred Comparisons:
- Package: Movie comparison unavailable
  A package is open, but no movie is loaded, so declared-vs-observed media checks were skipped.
```

How to read this sample

Sample result	Meaning	Operator next step
Readiness: 10 warnings	The package has warnings but no blockers in this sample.	Review each warning and determine whether package metadata should be corrected before delivery.
Next: Metadata / Declared ISAN is missing	Slate chooses a first actionable finding.	Jump to package metadata or inspect the source manifest for the missing ISAN.
Metadata warnings	ISAN, original spoken language, copyright,	Update package metadata or confirm the

Sample result	Meaning	Operator next step
	country, platform data, production company, cast, crew, and poster art are missing or unavailable in the package session.	omission is acceptable under your delivery rules.
Movie comparison unavailable	The package is open, but no movie was loaded; declared-vs-observed media checks were skipped.	Open the declared primary movie in the app, or provide observed-state to the CLI for scripted comparison.

Machine-readable analyze output

Use analyze output when tooling needs to inspect each finding. The command returns a JSON array. Each object includes category, severity, title, scope, code, evidence, identitySource, and fallbackUsed.

```
./slate analyze --package source-chapters-crop.itmsp/metadata.xml
[
  {
    "category" : "metadata",
    "evidence" : "The open package session does not declare Package.isan.",
    "fallbackUsed" : false,
    "code" : "metadata.declared_isan_is_missing",
    "scope" : "package-metadata",
    "title" : "Declared ISAN is missing",
    "identitySource" : "rule-contract",
    "severity" : "warning"
  }
  ... 9 additional findings omitted ...
]
```

Analyze field	Sample value	How to use it
category	metadata	Group findings by metadata, package, track, chapter, or other areas.
severity	warning	Drive pass/warning/blocker decisions.
title	Declared ISAN is missing	Display a human-readable finding name.
scope	package-metadata	Locate the finding domain, such as package-metadata or package-session.
code	metadata.declared_isan_is_missing	Use as a stable automation key.
evidence	The open package session does not declare Package.isan.	Show why the rule fired.
identitySource	rule-contract	Identify how the finding identity was produced.
fallbackUsed	false	Detect whether fallback logic was used.

Structured report output

Use report output when you need both operator-facing text and structured fields in one JSON document. In the sample, status is warning, schemaVersion is 1, the summary has 10 total findings, and nextFinding points to the missing ISAN warning.

```
./slate report source-chapters-crop.itmsp/metadata.xml
{
  "status" : "warning",
  "schemaVersion" : "1",
  "summary" : {
    "blockers" : 0,
    "total" : 10,
    "warnings" : 10
  },
  "operatorText" : "Summary:\n- Blockers: 0\n- Warnings: 10 ...",
  "nextFinding" : {
    "category" : "metadata",
    "code" : "metadata.declared_isan_is_missing",
    "scope" : "package-metadata",
    "title" : "Declared ISAN is missing",
    "severity" : "warning"
  },
  "findings" : [ ... ],
  "validationResultPayload" : {
    "schemaVersion" : "1",
    "findings" : [ ... ]
  }
}
```

Report field	Purpose	Sample value
status	Overall report state.	warning
schemaVersion	Report schema version.	1
summary	Compact counts for dashboards or gates.	blockers=0, warnings=10, total=10
operatorText	Human-readable summary suitable for logs or UI display.	Contains Summary, Next, Warnings, and Deferred Comparisons sections.
nextFinding	First recommended finding to inspect.	Declared ISAN is missing
findings	Full typed finding list.	10 finding objects in the sample.
validationResultPayload	Payload form for systems that only need code, scope, evidence, and fallbackUsed.	Includes schemaVersion and findings.

Sample findings table

Severity	Category	Title	Scope	Code
warning	metadata	Declared ISAN is missing	package-metadata	metadata.declared_isan_is_missing
warning	metadata	Declared original spoken language is missing	package-metadata	metadata.declared_original_spoken_language_is_missing
warning	metadata	Declared package copyright is missing	package-metadata	metadata.declared_package_copyright_is_missing
warning	metadata	Declared package country is missing	package-metadata	metadata.declared_package_country_is_missing
warning	metadata	Declared platform data is missing	package-metadata	metadata.declared_platform_data_is_missing
warning	metadata	Declared production company is missing	package-metadata	metadata.declared_production_company_is_missing
warning	metadata	Primary locale cast is missing	package-metadata	metadata.primary_locale_cast_is_missing
warning	metadata	Primary locale crew is missing	package-metadata	metadata.primary_locale_crew_is_missing
warning	metadata	Primary locale poster art is missing	package-metadata	metadata.primary_locale_poster_art_is_missing
warning	package	Movie comparison unavailable	package-session	package.movie_comparison_unavailable

Automation patterns

- Use readiness when a person needs to read the result in a terminal.
- Use analyze when a script needs to filter, count, or route individual findings.
- Use report when another system needs both a summary and the full finding payload.
- Gate releases on blockers first. Treat warnings according to your delivery policy.
- When package-only checks report deferred comparisons, add the movie in the app or supply an observed-state payload to enable declared-vs-observed media rules.

```
# Save a structured report for archival or CI processing.
./slate report source-chapters-crop.itmsp/metadata.xml > slate-report.json
```

```
# Example jq checks after the JSON is saved.
jq '.summary' slate-report.json
jq -r '.nextFinding.title' slate-report.json
jq '["findings[] | select(.severity == "warning")] | length' slate-report.json
```

8. Menu and Shortcut Reference

Action	Menu path	Shortcut
Open Movie	File > Open Movie...	Command-O
Open Package	File > Open Package...	Shift-Command-P
Open Recent	File > Open Recent	-
Add Reference Track	Track > Add Reference Track...	-
Delete Reference Track	Track > Delete Reference Track	Command-D
Show Chapter View	View > Chapters	Control-Command-C
Show Track View	View > Tracks	Control-Command-T
Show Package View	View > Package	Control-Command-P
Toggle Subtitles	View > Show Subtitles	-
Toggle Closed Captions	View > Show Closed Captions	-
Toggle Autoload Sidecars	View > Autoload Sidecars	-
Enter Full Screen	View > Enter Full Screen	Control-Command-F
Show Package Summary	Validation > Show Package Summary	Option-Command-P
Show Readiness Report	Validation > Show Readiness Report	Option-Command-R

9. Troubleshooting

Problem	Likely cause	What to try
No movie is visible	Unsupported or unplayable movie file.	Confirm the file is .mov or .mp4. If Slate reports that the movie format is not supported for playback, the file may be readable for metadata but not playable through the current macOS media stack.
Package checks seem incomplete	The package is open but no matching movie is loaded.	Open the package first, then open the declared primary movie. Confirm the movie path matches the package declared primary video asset.
Subtitle or caption controls are disabled	No matching sidecar track is present or enabled.	Add a supported sidecar with Track > Add Reference Track.... Confirm sidecar autoload is enabled and sidecar files share the movie base filename.
Readiness links do not jump where expected	Some findings point to package-level state rather than a single row.	Open the relevant pane, then try the readiness link again.
CLI reports deferred movie comparison	The CLI has package data but no observed movie state.	Open the declared movie in the app, or supply an observed-state JSON payload for scripted checks.
Runtime-missing message appears	Runtime helpers are not installed next to Slate.app.	Reinstall or rebuild the app bundle so SlatePackageRuntime, SlateTrackRuntime, SlateChapterRuntime, SlateReviewRuntime, SlateTimelineRuntime, and SlateValidationRuntime are present.

10. Operator Notes

- Slate preserves recent operator inputs and workspace selections between launches.
- Pane widths, selected modes, and some inspector preferences are remembered.
- A package-only readiness result is still useful for metadata review, but movie-aware findings require either an opened movie in the app or an observed-state payload in CLI workflows.
- Keep package and movie paths stable during review so recent inputs and readiness links remain meaningful.

Appendix A. Public Application Build Notes

These build notes are retained for developers and release operators. They are separated from the operator workflow so new users can learn the product without reading build configuration first.

Public-facing Slate.app adapter package

This package contains the Xcode project, public adapter source, paneharness target, and sibling runtime artifacts needed to build and run the public Slate.app surface.

paneharness.app loads adapter classes from the sibling Slate.app. The public paneharness target depends on Slate, so building or running the harness from Xcode also builds Slate.app into the same build directory.

Build

From this directory:

```
./build_public_apps.sh
```

The script builds the public Slate.app adapter and paneharness.app in Release configuration and writes both products to build/Release. It uses generic/platform=macOS so it does not depend on Xcode selected toolbar destination.

build/Release is local build output and is not part of the public repository. The runtime binaries live in Runtime/; the Xcode targets copy those binaries next to Slate.app and paneharness.app during each local build.

Xcode destination

If Xcode shows Please select an available device or choose a simulated device as the destination when you click Run, choose a concrete macOS destination:

```
Product > Destination > My Mac
```

You can also use the destination picker next to the scheme selector and choose My Mac. This project builds macOS apps; it does not need an iOS simulator. Xcode stores the selected run destination as local user interface state, so the public project does not check that setting into source control.

paneharness run configuration

paneharness.app loads adapter classes from a sibling Slate.app. In this public package the useful build products live together under build/Release. The shared harness target builds Slate.app first and lets the Slate target copy the sibling runtime tools.

If paneharness opens an alert saying Slate.app must be in the same directory as paneharness.app and the expected path contains build/Debug/Slate.app, Xcode is running the harness with a Debug Run configuration. Change it with:

```
Product > Scheme > Edit Scheme... > Run > Info > Build Configuration > Release
```

The shared paneharness scheme is set this way, but Xcode may keep local scheme state if the project was already open.

Glossary

Term	Meaning
Blocker	A readiness finding that should be corrected before delivery.
Warning	A readiness finding that may require operator review.
Package	A JSON package file or Apple manifest XML file containing declared delivery metadata.
Observed state	The media and UI state Slate can see from the currently opened movie and session, or a JSON payload supplied to CLI workflows.
Readiness finding	A pass, warning, or blocker produced by a readiness rule.
Reference track	A temporary audio, subtitle, caption, or movie-derived track added to the current session for inspection.
Sidecar	An external subtitle or caption file associated with the movie.
Readiness rail	The pane-level findings area that summarizes and links to relevant issues.